

Blood Pressure Levels and Outcomes in Type 2 Diabetes

Dose-Response Meta-Analysis of 5.87 Million Cohort Participants

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ABSTRACT

BACKGROUND The dose-response relationships between systolic and diastolic blood pressure (SBP and DBP) levels and risks of all-cause mortality, cardiovascular and renal outcomes in type 2 diabetes remain poorly characterized, particularly at lower blood pressure (BP) levels.

OBJECTIVE This study aimed to examine risk patterns of major clinical outcomes across a wide range of BP levels in participants with type 2 diabetes.

METHODS We systematically searched PubMed, Embase, and Web of Science from inception to November 30, 2024, for cohort studies assessing associations of BP levels with all-cause mortality, cardiovascular, and renal outcomes. One-stage mixed-effects dose-response meta-analysis was conducted to assess the curvilinear associations.

RESULTS A total of 89 cohorts from 113 articles of 5,875,364 participants with type 2 diabetes were identified. A J-shaped association was observed for SBP with all-cause mortality. J-shaped associations were also observed for SBP with cardiovascular events and for DBP with all-cause mortality, with flattened risks at lower BP levels. After excluding studies with participants who had baseline cardiovascular diseases or cancer, a significantly lower risk of cardiovascular events was observed at lower SBP levels, as well as a flattened risk of all-cause mortality. Positive linear or monotonic trends were observed for renal events, estimated glomerular filtration rate decline, and development or progression of albuminuria.

CONCLUSIONS In type 2 diabetes, BP is linearly or monotonically associated with most cardiovascular and renal outcomes. Low SBP does not appear to be associated with an elevated risk of all-cause mortality compared with higher BP targets, suggesting that previously reported associations are likely due to reverse causation and unmeasured confounding. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS
AND ACRONYMS****BP** = blood pressure**CAD** = coronary artery disease**DBP** = diastolic blood pressure**eGFR** = estimated glomerular
filtration rate**ESRD** = end-stage renal
disease**SBP** = systolic blood pressure**T2D** = type 2 diabetes

Hypertension, with an age-standardized global prevalence of 32% in women and 34% in men aged 30 to 79 years, is the leading cause of cardiovascular and renal diseases worldwide.¹ Untangling the dose-response relationships between blood pressure (BP) and various clinical outcomes can help tailor current prevention and treatment strategies.

Despite mounting clinical evidence supporting the benefits of BP reduction, the optimal BP target for patients with diabetes remains uncertain, and current guidelines offer discrepant recommendations.²⁻⁴ This issue cannot be adequately addressed without a comprehensive evaluation of risk patterns across the full spectrum of BP levels. Although randomized controlled trials (RCTs) provide criterion-standard evidence for causal inference, they are inherently limited to predefined treatment targets and are underpowered to examine the risk patterns at the extremes of BP distribution. Observational studies may be advantageous over clinical trials in terms of population generalizability and the capacity to evaluate less frequent outcomes with longer durations of follow-up. However, single-cohort studies have reported inconsistent findings regarding the dose-response relationships between BP and clinical outcomes in patients with diabetes, and less evidence is currently available for certain outcomes, such as renal events. Furthermore, the limited sample size at the lower end of the BP distribution in cohort studies has left a large gap regarding the dose-response relationships of lower BP levels with clinical outcomes.

Here, we performed a 1-stage dose-response meta-analysis using cohort studies from multiple continents to examine the associations of systolic and diastolic blood pressure (SBP and DBP) levels with the risks of all-cause mortality, cardiovascular, and renal outcomes in participants with type 2 diabetes. By analyzing all relevant cohort data reported to date, we aimed to assess the risk patterns of major clinical outcomes across a wide range of BP levels, as empowered by a huge sample size, a wide coverage of study populations, and a long duration of follow-up.

METHODS

SEARCH STRATEGY AND SELECTION CRITERIA. The conduct and reporting of this systematic review and meta-analysis followed the Cochrane Handbook for Systematic Reviews⁵ and Preferred Reporting Items for Systematic Reviews and Meta-Analyses⁶ guidelines. The original protocol was prospectively

registered in PROSPERO (Blood Pressure Levels and Risks of Clinical Outcomes in Patients With Type 2 Diabetes: A Systematic Review and Meta-analysis of Cohort Studies; [CRD42022370546](#)). Because this was a meta-analysis of previously published and ethically approved studies, additional ethical review was not required.

Two investigators independently and in duplicate searched PubMed, Embase, and Web of Science using terms provided in [Supplemental Table 1](#) for cohort studies conducted in human adults and published in English-language journals from inception to November 30, 2024. References of included articles and related meta-analyses were manually searched to identify additional studies.

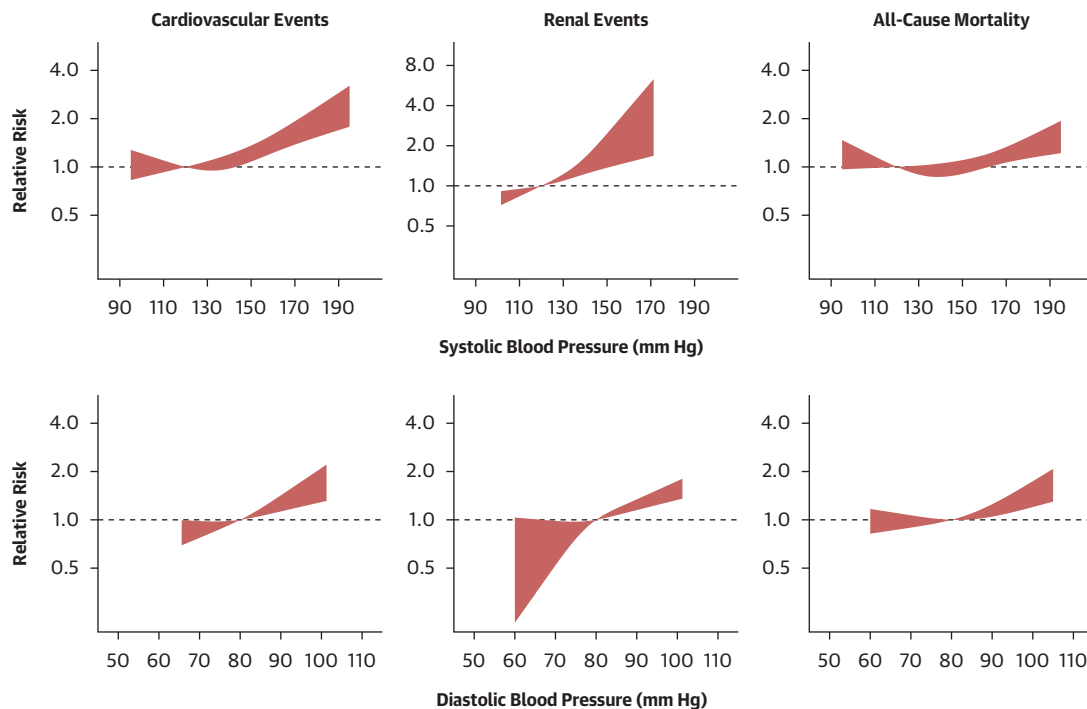
Cohort studies that had ≥ 100 participants with type 2 diabetes and a follow-up duration of ≥ 1 year, and that reported associations of SBP or DBP levels with clinical outcomes including cardiovascular outcomes, renal outcomes, or all-cause mortality in people with type 2 diabetes were included ([Central Illustration](#)). Detailed exclusion criteria are listed in [Supplemental Table 2](#). If there was >1 eligible article from the same cohort, the one with the most complete information on BP levels and outcomes of interest, and with the greatest person-years of follow-up was used. When the follow-up person-years were not provided, the number of participants with type 2 diabetes was multiplied by the follow-up duration to make an approximate estimation.

DATA EXTRACTION AND SYNTHESIS. Two investigators independently and in duplicate extracted data on cohort name (or study site), study design (prospective or retrospective), country or region, number of participants with type 2 diabetes, follow-up duration (years), age (range), proportion of female participants, use of BP-lowering medications, and prevalence of comorbidities such as cardiovascular diseases and cancer at baseline.

Clinic BP values were prioritized. If unavailable, home BP or 24-hour ambulatory BP values were used. For articles reporting BP categories, the mean value for each category was documented. If the mean value for participants with type 2 diabetes was unavailable, the mean value from the general population was used instead; otherwise, the mean of the interval bounds was calculated and used. For open-ended intervals, the interval width was estimated as 1.25 times the adjacent interval.⁷ We also recorded the number of events and participants within each BP category. If either was missing, we estimated the missing data by using known variables and effect estimates for each BP category.⁸ If both were missing, we approximated

CENTRAL ILLUSTRATION Framework of This Dose-Response Meta-Analysis**Overview of the Current Dose-Response Meta-Analysis**

	Population	5,875,364 participants with type 2 diabetes
	Exposure	Systolic and diastolic blood pressure levels
	Primary Outcomes	Cardiovascular events, renal events, and all-cause mortality
	Study Design	Dose-response meta-analysis of 89 cohort studies
	Search Date	From inception to November 30, 2024

Blood Pressure Levels and Risks of Primary Outcomes

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(Top) Overview of the current dose-response meta-analysis based on the PICOS framework. (Bottom) Curvilinear dose-response associations between blood pressure levels and the risks of primary outcomes. The red shaded areas represent 95% CIs.

the participants' distribution with the use of real-world diabetes cohorts or national statistics, assuming that the missing data followed the same distribution as observed in those data sources. The assumptions and approaches for estimation are described in [Supplemental Methods 1](#), and studies with imputed number of participants or events are listed in [Table 1](#).

For articles reporting continuous BP values, BP increment units were recorded along with the total

number of events and participants. We prioritized the increment unit of 10 mm Hg for SBP and 5 mm Hg for DBP. If these increments were not reported, the BP intervals used in the original articles (eg, 15 mm Hg) were documented instead.

Relative risk (RR) with a 95% CI was extracted as the effect estimate in this meta-analysis. Graphic data of the effect estimate were extracted by an online Plot Digitizer tool. Effect estimate other than RR was converted to RR using the following

TABLE 1 Cohorts With Imputed Participant or Event Data

PMID	Cohort With Imputation	Data Source Used for Imputation	PMID
27339442	National Taiwan University Hospital	Taiwan Diabetes Study (TDS)	29458073
33288879	KAMOGAWA-HBP study	Japan Medical Data Center Claims Database	34035075
36732668	National Database of Health Checkups in Japan	Japan Medical Data Center Claims Database	34035075
		Japan National Health and Nutrition Survey (2018)	-
38671217	Japan Medical Data Center Claims Database	Japan Medical Data Center Claims Database	34035075
		Japan National Health and Nutrition Survey (2010)	-
25512005	Clinical Practice Research Datalink	UK Biobank	-
25645749	German Health Fund AOK PLUS	UK Biobank	-
27402917	Western Denmark Heart Registry	UK Biobank	-
34315938	Madrid Diabetes Study (MADIABETES)	UK Biobank	-
29929484	Kaiser Permanente Northern California	Kaiser Permanente Southern California	25104529
		National Health and Nutrition Examination Survey (2009-2010)	-

methods: 1) HR was extracted and treated as RR⁹; 2) incident rate was applied to calculate RR by dividing the rate of the target BP category by that of the reference BP category, with 95% CI estimated using the number of events and participants in the reference and also the target categories^{10,11}; 3) OR was converted to RR according to a corrected method by Zhang and Yu,¹² with the total event rate used as an alternative if the reference group's event rate was unavailable; and 4) beta coefficient was first transformed into HR or OR according to the regression model used in the original article, and was subsequently converted to RR by means of the methods described above.^{13,14} The assumptions and formulas used for converting these effect estimates to RRs are described in [Supplemental Methods 2](#). For studies reporting floated CIs estimated by the floating absolute risk method, the floated CIs were converted to conventional CIs following the approach of Orsini.¹⁵

When data were reported by subgroups (eg, sex), effect estimates within the same BP category were pooled with the use of the fixed-effects model. For articles reporting BP categories, the group with the lowest BP value was used as the reference group. In the absence of a reference group, we assigned the group with the lowest BP value as the reference, and calculated the 95% CI using the method by Katz et al.¹¹ The Hamling approach was applied when the lowest BP group was not used as the reference in the original study.¹⁶

PRIMARY AND SECONDARY OUTCOMES. Primary outcomes included: 1) all-cause mortality; 2) cardiovascular events; and 3) renal events. We defined cardiovascular events as a composite of at least 2 of the following: coronary artery disease (CAD), heart failure, stroke, peripheral artery disease, or

cardiovascular mortality. We prioritized major adverse cardiovascular events (MACE), defined as a composite of myocardial infarction, stroke, or cardiovascular mortality, with hospitalization for unstable angina or heart failure as additional optional components.¹⁷ If MACE data were not available, we used the outcome comprising the largest number of cardiovascular components. We defined renal events as a composite of at least 2 of the following: development or progression of albuminuria, doubling of serum creatinine, decline in estimated glomerular filtration rate (eGFR), end-stage renal disease (ESRD), or death due to renal diseases. We prioritized a modified definition of major adverse kidney events (MAKE), defined as 2 or more of the following: renal dysfunction (assessed by means of serum creatinine or eGFR), dialysis or renal transplantation, or death due to renal diseases.¹⁸ If MAKE was not available, we used the outcome comprising the largest number of renal components.

Secondary outcomes included: 1) CAD; 2) stroke; 3) cardiovascular mortality; 4) eGFR decline; 5) development or progression of albuminuria; and 6) ESRD.

QUALITY ASSESSMENT. Two investigators independently and in duplicate assessed the quality of each included article using the Risk Of Bias in Non-randomized Studies - of Exposures (ROBINS-E) tool.¹⁹ In addition, the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) tool was applied to evaluate the certainty of evidence for the traditional pairwise meta-analysis.²⁰ Detailed descriptions of each domain and the grading criteria for the GRADE tool are presented in [Supplemental Methods 3](#).

STATISTICAL ANALYSIS. Curvilinear dose-response associations. The curvilinear dose-response association was measured by means of the 1-stage mixed-

effects approach established by Orsini.²¹ Compared with the traditional 2-stage method, this approach pools aggregate data before trend analysis and estimates more flexible curves through a mixed-effects model. It allows for the inclusion of studies with only 2 categories, thereby maximizing evidence integration.²²

Restricted cubic splines using 3, 4, and 5 knots were compared for model fit with knot locations following Harrell's recommendation and both tails restricted to being linear.²³ The maximum likelihood estimation method and the Akaike information criterion were used to determine the best-fitting curve. The significance of nonlinearity was assessed by testing the regression coefficients of the spline (except the first coefficient) equal to zero, with $P < 0.05$ indicating departure from linearity. In addition, a Wald-type test for the hypothesis of no exposure-disease association was performed.

Monotonic or linear trends were used to describe consistently increasing risk patterns, depending on the curve's shape and the P value for nonlinearity. We referred to a trend as "monotonic" when the P value for nonlinearity was <0.05 , which suggested that a nonlinear model provided a better fit than a linear one but the increasing trend of the curve was consistent. If the estimated RR increased at lower BP levels, we assessed whether the association followed a J-shaped or U-shaped pattern by examining the predicted RRs along the fitted curve and changes in its gradient (defined as the change in RR per 1-mm Hg increase in BP), as visualized in scatter plots. We considered a curve to be U-shaped if it was symmetric with risks significantly increased at both lower and higher ends, in which case the gradient changed from <0 (negative association) to >0 (positive association) in an approximately linear manner as BP levels increased. If no significantly increased risk was observed at the lower end, a J-shaped curve was identified. In addition, if the gradient showed a relatively flat pattern near the value of 0 (RR changes very little) and then increased steeply to above 0 (positive association) with increasing BP levels, we described it as a J-shaped curve with a flattened lower end.

To minimize the impact of extreme BP estimates, the ranges for SBP and DBP were set at 90 to 200 mm Hg and 50 to 110 mm Hg, respectively, with the out-of-range values within each study excluded. The reference values of SBP and DBP for the curvilinear dose-response analysis were set at 120 mm Hg and 80 mm Hg, respectively. The estimated RRs and corresponding 95% CIs were recorded using 5-mm Hg intervals along the curve.

Several sensitivity analyses were conducted. To assess the potential impact of reverse causation on RRs at lower BP levels, we excluded studies with participants who had baseline cardiovascular diseases or cancer, as identified from baseline characteristics or the inclusion and exclusion criteria of each study. To evaluate the potential impact of BP measurement method on the outcomes, we excluded studies with out-of-office BP measurements. To evaluate the impact of data imputation, we excluded studies with imputed number of participants or events. To assess potential small-study effects, we excluded studies with $<1,000$ participants. We also evaluated the influence of extremely large studies by excluding the largest study for each outcome. Regarding effect estimates, we excluded studies that reported ORs or beta coefficients from logistic regression models, owing to the potential overestimation during their conversion to RRs. Regarding study quality, we excluded studies with an overall high or critical risk according to the ROBINS-E tool, or studies with a retrospective design.

Risks of outcomes with continuous BP increment. In addition to the curvilinear dose-response analysis, we also conducted the traditional pairwise meta-analysis to examine the risks of clinical outcomes in associations with continuous BP levels by pooling RRs per 10-mm Hg increment of SBP or per 5-mm Hg increment of DBP. The pairwise meta-analysis was not conducted for U- or J-shaped associations (except for those with a flattened lower end) found in the curvilinear dose-response analysis, to limit the analysis for unidirectional risk associations. When both categorical and continuous BP data were available from the same cohort, we used the continuous BP data to avoid bias from data transformation. When only categorical BP data were reported, we conducted a linear dose-response analysis to estimate risks of outcomes to maximize data utilization and increase the mathematical tractability of the analysis.²⁴ For studies using another increment unit (eg, 15 mm Hg), RRs and 95% CIs were standardized to the 10-mm Hg unit for SBP and 5-mm Hg unit for DBP by the method of Taylor et al.²⁵

To account for the anticipated high heterogeneity and the relatively small sample size for certain secondary outcomes, we performed the pairwise meta-analysis using a random-effects model, with 95% CIs for the pooled RRs estimated by the Hartung-Knapp-Sidik-Jonkman method.²⁶ The heterogeneity variance τ^2 (with 95% CI) was estimated by means of the restricted maximum likelihood method. Additional measures of heterogeneity included the I^2 statistic with 95% CI, the H statistic, and the P value

from Cochran Q-test.²⁷ As a sensitivity analysis, we additionally applied the conventional DerSimonian and Laird model.²⁸

We conducted subgroup analysis to assess the consistency of the associations of BP levels with primary outcomes across categorized baseline variables, including mean (or median) baseline age, female proportion, geographic location, study design, number of participants with type 2 diabetes, follow-up duration, and the overall risk of bias as assessed by the ROBINS-E tool. Meta-regression analysis was conducted to examine differences between subgroups. When sufficient data were available (≥ 10 studies), the meta-regression model was further used to evaluate whether continuous or ordinal baseline variables contributed to heterogeneities in the associations between BP levels and outcomes.²⁹

Assessment of publication bias was limited to the analysis of continuous BP levels. Funnel plots were assessed, together with the Egger's test when sufficient data were available (≥ 10 studies).³⁰ Owing to the potential inflation of type I error in the Egger's test when heterogeneity is present, we additionally conducted the arcsine-Thompson test to detect potential publication bias.³¹ RRs were adjusted by the trim-and-fill method if the *P* value for either test was < 0.05 .³²

Stata/MP version 17.0 (StataCorp) and R version 4.3.2 (R Foundation for Statistical Computing) were used for all statistical analyses. Two-sided *P* < 0.05 was considered to be significant.

RESULTS

A total of 89 cohorts from 113 articles, involving 5,875,364 participants with type 2 diabetes, were included in this meta-analysis (Figure 1, Supplemental Table 3, Central Illustration). The mean follow-up duration was 6.3 years, ranging from 1 to 18 years. The average age was 60.5 years. Among the cohorts, 57 (64.0%) reported a prospective study design, and 1 (1.1%) included only male participants (Supplemental Table 4); 84 cohorts (94.4%) reported office BP, 3 (3.4%) reported ambulatory BP, 1 (1.1%) reported home BP, and 1 (1.1%) reported mixed office and home BP (Supplemental Table 5). Definitions of outcomes and corresponding model adjustments are summarized in Supplemental Table 6. Of the 113 articles, 82 were classified as having low risk, 18 moderate risk, 6 high risk, and 7 critical risk (Supplemental Table 7).

CURVILINEAR DOSE-RESPONSE ASSOCIATIONS.

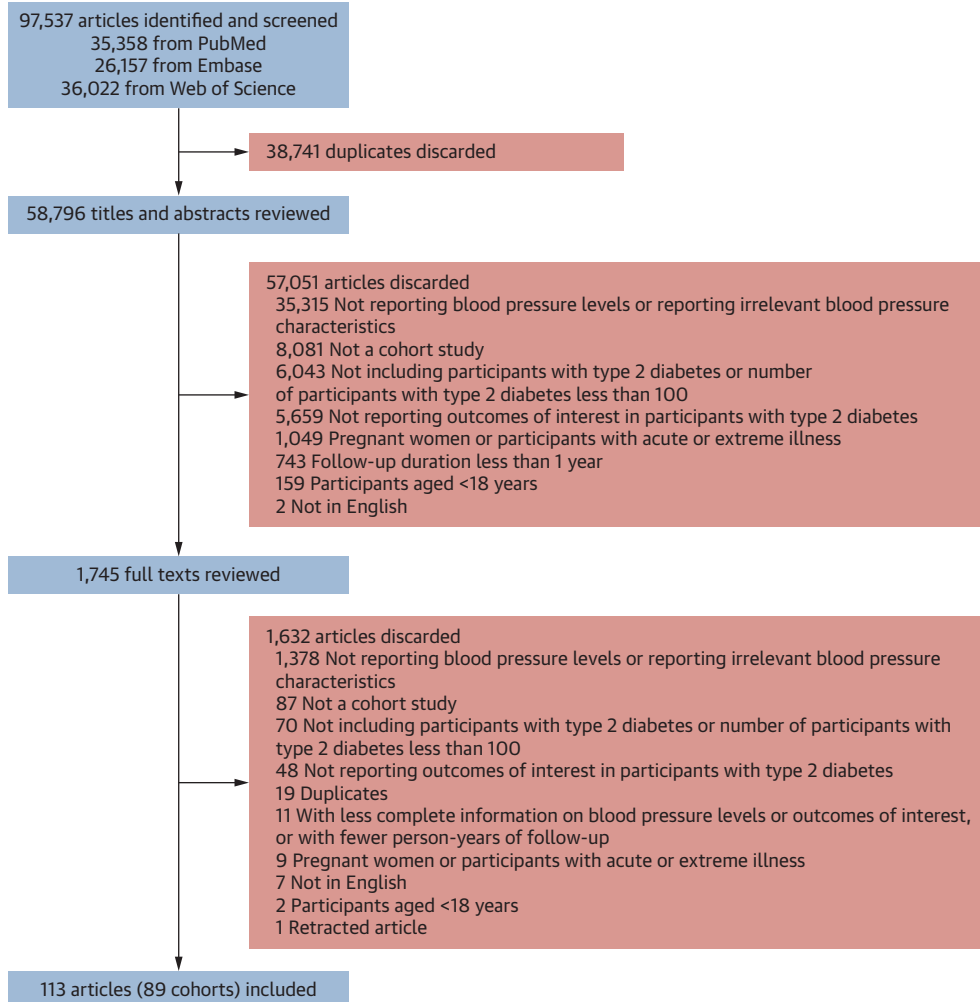
The 3-knot curvilinear model with knots at the 10th,

50th, and 90th percentiles achieved convergence for all the associations and showed generally better model fit than 4-knot and 5-knot models (Supplemental Table 8), and therefore it was used for analysis. The numbers of participants and events in each 10-mm Hg SBP category are presented in Supplemental Table 9. Overall, a large number of participants with SBP < 120 mm Hg were included for most of the examined outcomes. For example, approximately 1.12 million out of 4.64 million participants (24.1%) were in SBP categories < 120 mm Hg for all-cause mortality. A J-shaped association was observed for SBP and all-cause mortality (Figure 2, Supplemental Table 10, Supplemental Figure 1). Meanwhile, we detected a J-shaped association for SBP and cardiovascular events, with a flattened risk at lower SBP levels. In addition, a linear association was observed between SBP and renal events.

A large number of participants with DBP < 70 mm Hg were included for most of the examined outcomes (Supplemental Table 11). However, participants with DBP < 60 mm Hg were reported only for cardiovascular mortality. We detected a J-shaped association between DBP and all-cause mortality, with a flattened risk at lower DBP levels (Figure 2, Supplemental Table 12, Supplemental Figure 1). Linear associations were observed between DBP and both cardiovascular and renal events.

For secondary cardiovascular outcomes, a linear trend was observed between SBP and cardiovascular mortality (Figure 3, Supplemental Tables 9 and 10). Although no statistically significant lower risk was observed for CAD or stroke at SBP < 120 mm Hg, both outcomes exhibited a monotonic trend with SBP levels. Similar results were observed for DBP levels and secondary cardiovascular outcomes (Figure 3, Supplemental Tables 11 and 12). All secondary renal outcomes were associated with BP levels in a linear or positive monotonic pattern (Figure 4, Supplemental Tables 9 to 12). Analysis was not conducted for DBP and albuminuria or for DBP and ESRD owing to a lack of significant dose-response association ($P_{\text{dose-response}} > 0.05$) or insufficient number of eligible cohorts.

After excluding studies with participants who had baseline cardiovascular diseases or cancer, there were still a large number of participants with SBP lower than 120 mm Hg, especially for all-cause mortality ($n = 554,120$) (Supplemental Table 13). A monotonic trend was found between SBP levels and cardiovascular events, with lower SBP levels associated with significantly lower risks (Figure 5,

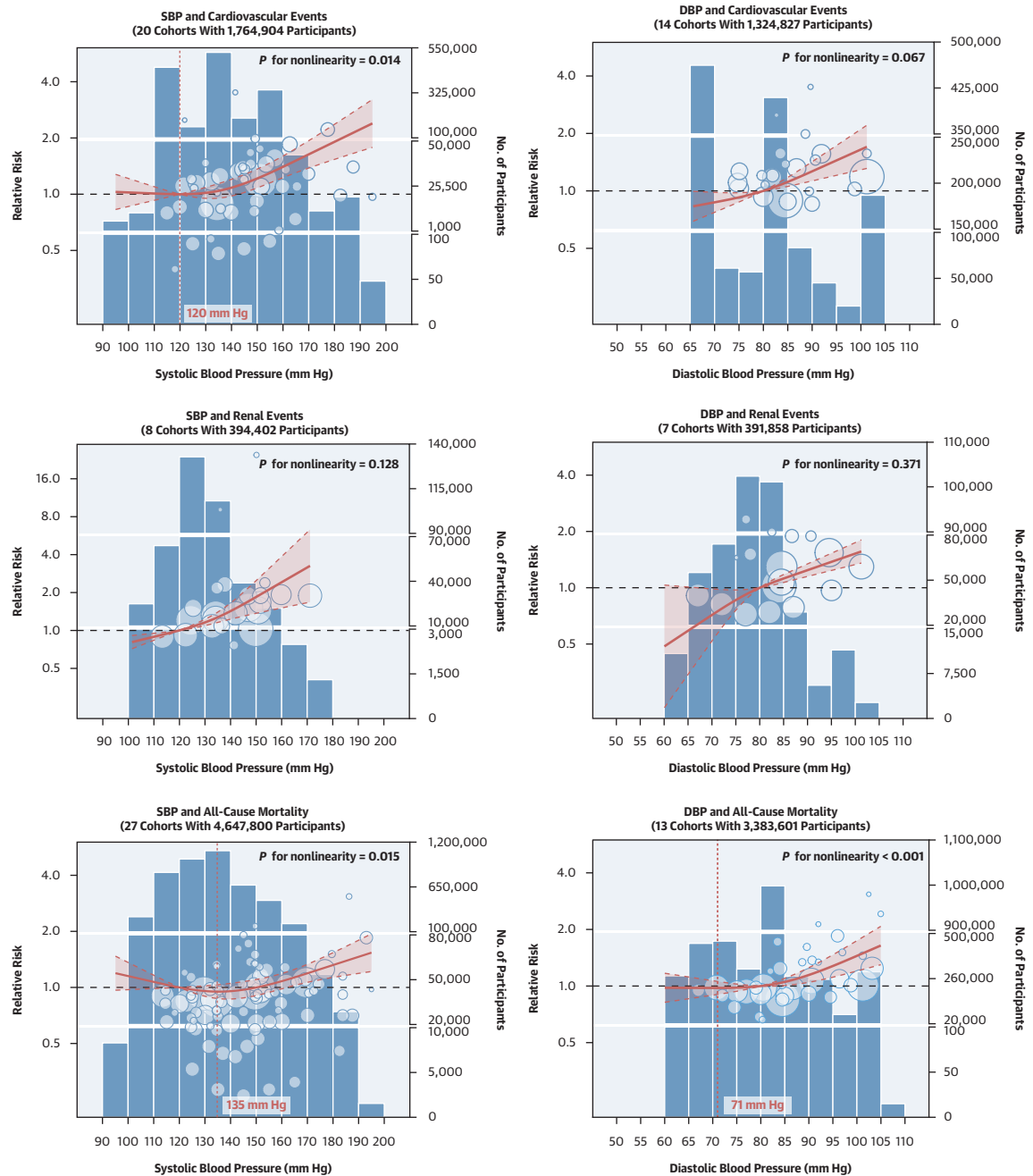
FIGURE 1 Flow Chart of Included Cohorts

Detailed inclusion and exclusion criteria are presented in [Supplemental Table 2](#). A total of 89 cohorts from 113 articles were finally included in this meta-analysis.

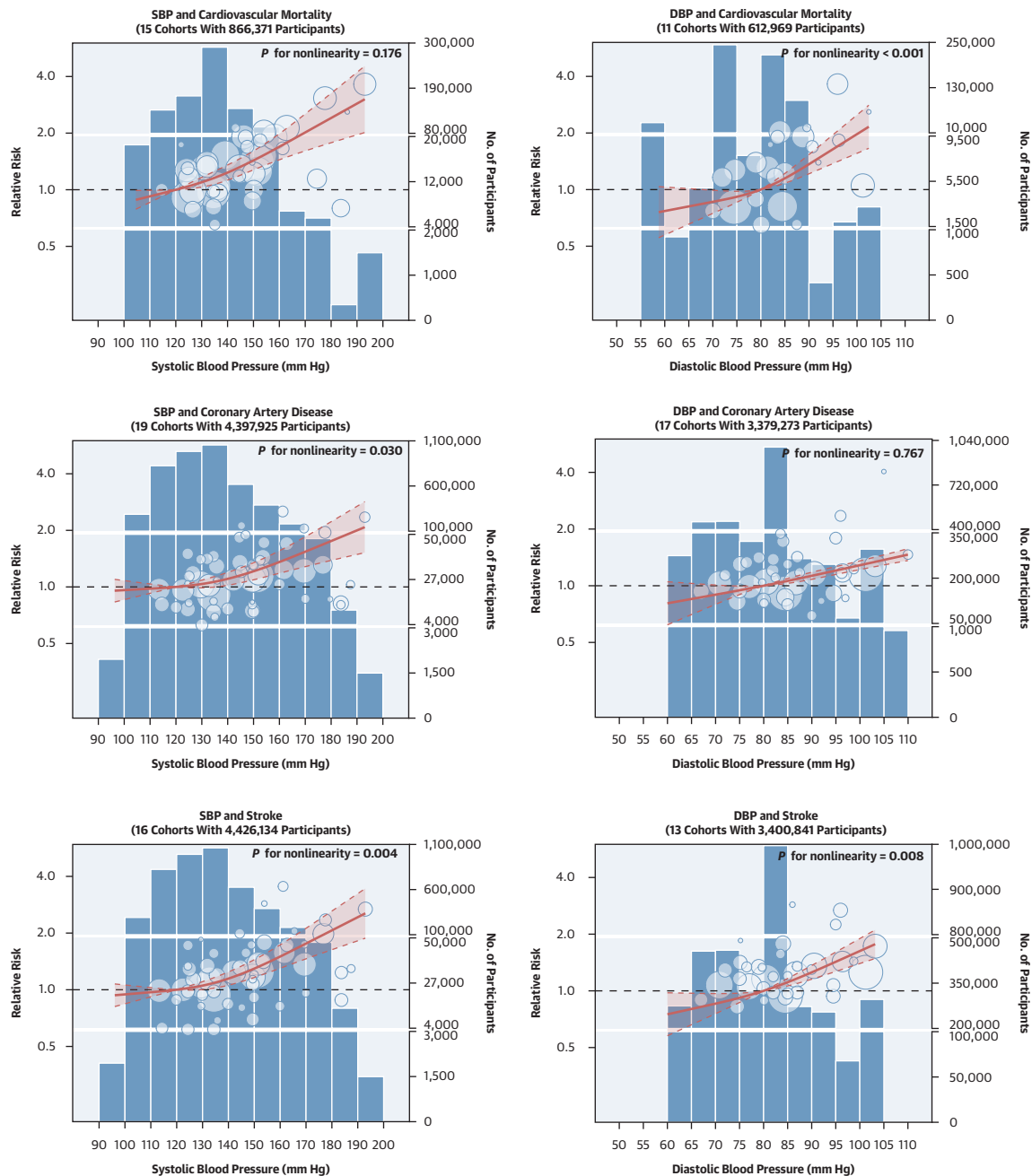
[Supplemental Table 14](#)). In addition, we observed a J-shaped curve with a flattened lower end for SBP and all-cause mortality. A monotonic risk pattern was detected for DBP and all-cause mortality ([Figure 5](#), [Supplemental Tables 15 and 16](#)). Results from other sensitivity analyses are shown in [Supplemental Figures 2 to 16](#). Most associations were similar to those of the main analysis.

RISKS OF OUTCOMES WITH CONTINUOUS BP INCREMENT. We did not conduct the traditional pairwise meta-analysis for the association between SBP and all-cause mortality, but instead conducted a systematic review of cohorts reporting the

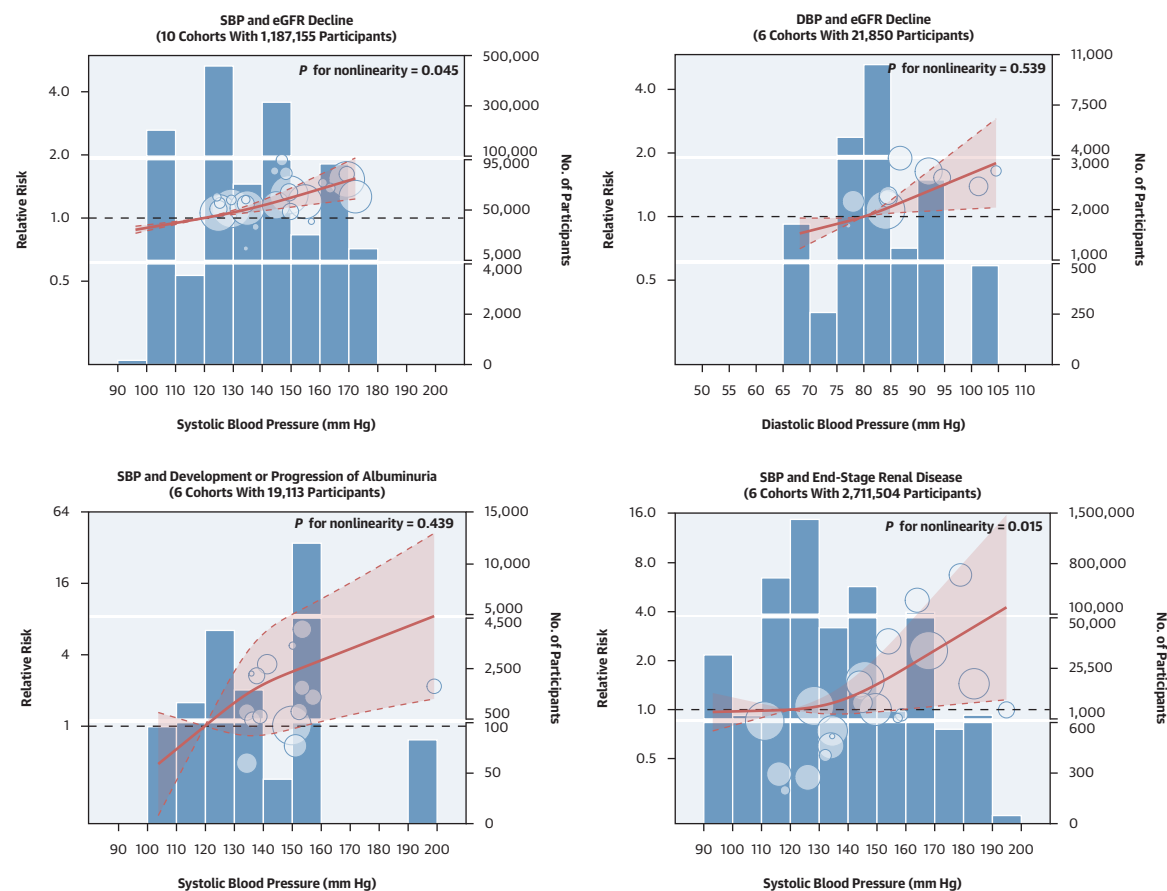
association ([Supplemental Table 17](#)). Results using the Hartung-Knapp-Sidik-Jonkman method indicated that each 10-mm Hg increase in SBP was associated with a 6% higher risk of cardiovascular events (RR: 1.06; 95% CI: 1.04-1.08), and a 22% higher risk of renal events (RR: 1.22; 95% CI: 1.11-1.34) ([Supplemental Table 18](#), [Supplemental Figures 17 and 18](#)). RRs for each 5-mm Hg increase in DBP were 1.04 (95% CI: 0.99-1.10) for all-cause mortality, 1.06 (95% CI: 1.02-1.10) for cardiovascular events, and 1.10 (95% CI: 1.00-1.22) for renal events ([Supplemental Table 18](#), [Supplemental Figures 19 to 21](#)). Secondary outcomes showed a similar trend of increasing risks associated with each 10-mm Hg increase in SBP or

FIGURE 2 Dose-Response Associations of BP Levels With Primary Outcomes

The red solid lines indicate relative risks, with the shaded areas representing 95% CIs. The red vertical lines indicate BP levels with outcome risks at nadirs for J-shaped curves. Each circle represents a BP category within a study, with the potential of multiple circles corresponding to different categories from the same study. Circle locations indicate relative risk point estimates across BP categories, with sizes inversely proportional to the standard error (not comparable across outcomes). Blue bars represent the numbers of participants within each 10-mm Hg SBP or 5-mm Hg DBP interval. DBP = diastolic blood pressure; SBP = systolic blood pressure.

FIGURE 3 Dose-Response Associations of BP Levels With Secondary Cardiovascular Outcomes

The red solid lines indicate relative risks, with the shaded areas representing 95% CIs. Each circle represents a BP category within a study, with the potential of multiple circles corresponding to different categories from the same study. Circle locations indicate relative risk point estimates across BP categories, with sizes inversely proportional to the standard error (not comparable across outcomes). Blue bars represent the numbers of participants within each 10-mm Hg SBP or 5-mm Hg DBP interval. DBP = diastolic blood pressure; SBP = systolic blood pressure.

FIGURE 4 Dose-Response Associations of BP Levels With Secondary Renal Outcomes

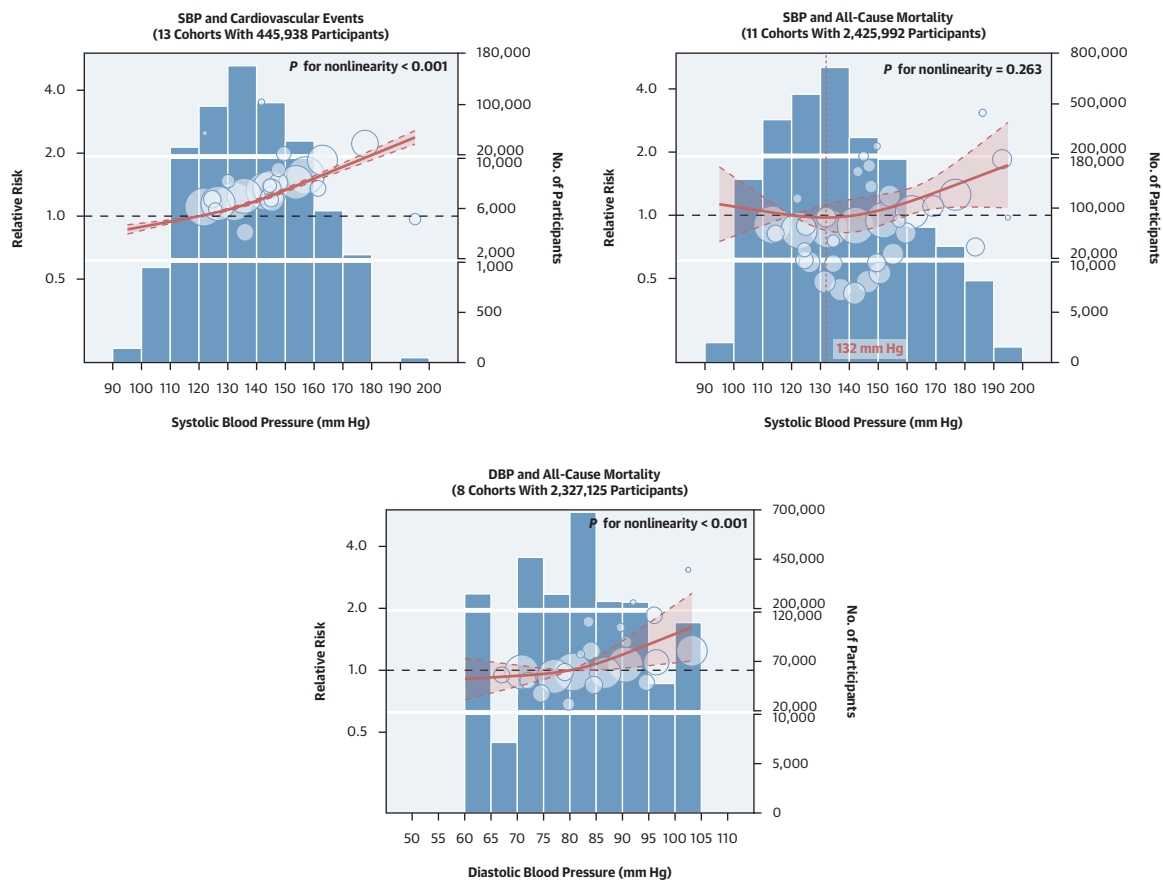
The red solid lines indicate relative risks, with the shaded areas representing 95% CIs. Each circle represents a BP category within a study, with the potential of multiple circles corresponding to different categories from the same study. Circle locations indicate relative risk point estimates across BP categories, with sizes inversely proportional to the standard error (not comparable across outcomes). Blue bars represent the numbers of participants within each 10-mm Hg SBP or 5-mm Hg DBP interval. DBP = diastolic blood pressure; SBP = systolic blood pressure.

5-mm Hg increase in DBP (Supplemental Table 18, Supplemental Figures 22-33). High heterogeneities were detected across outcomes. The DerSimonian and Laird model yielded similar results (Supplemental Table 18).

Leave-one-out analysis showed that the results were not substantially influenced by the exclusion of any single study (Supplemental Figures 34 to 50). Subgroup analysis showed a consistent direction of associations between BP levels and the primary outcomes, with reduced heterogeneities observed in some subgroups (Tables 2 and 3, Supplemental Tables 19 to 21). Meta-regression indicated that longer follow-up duration was associated with a greater SBP-related risk of cardiovascular events, explaining most of the heterogeneity (adjusted $R^2 = 99.54\%$) (Table 4). However, meta-regression

incorporating baseline characteristics and study quality failed to identify factors that could explain other observed heterogeneities (Table 4, Supplemental Table 22).

CERTAINTY OF EVIDENCE FOR PAIRWISE META-ANALYSIS. Funnel plots for each outcome are presented in Supplemental Figures 51 to 67. Owing to the insufficient number of eligible studies ($n < 10$), Egger's test and arcsine-Thompson test were not conducted for the associations between DBP and eGFR decline, development or progression of albuminuria, or ESRD. After adjustment for publication bias by means of the trim-and-fill method for outcomes with $P < 0.05$ in the Egger's test or arcsine-Thompson test, the direction of RRs remained consistent (Supplemental Table 23).

FIGURE 5 Sensitivity Analysis Excluding Participants With Baseline Cardiovascular Diseases or Cancer

This analysis was limited to outcomes reporting a J-shaped association with BP levels. The red solid lines indicate relative risks, with the shaded areas representing 95% CIs. The red vertical line indicates the BP level with outcome risk at the nadir for the J-shaped curve. Each circle represents a BP category within a study, with the potential of multiple circles corresponding to different categories from the same study. Circle locations indicate relative risk point estimates across BP categories, with sizes inversely proportional to the standard error (not comparable across outcomes). Blue bars represent the numbers of participants within each 10-mm Hg SBP or 5-mm Hg DBP interval. BP = blood pressure; DBP = diastolic blood pressure; SBP = systolic blood pressure.

According to the GRADE tool, all outcomes, except for DBP and albuminuria, were downgraded owing to inconsistency arising from heterogeneity. In addition, the certainty for SBP and renal events was downgraded owing to strongly suspected publication bias, and the certainty for DBP and all secondary renal outcomes was downgraded owing to a limited number of cohorts ($n < 10$). The certainty for several outcomes was upgraded owing to the large magnitude of effects or linear dose-response associations. As a result, the certainty of evidence was assessed as high for the association of SBP with cardiovascular events and for DBP with renal events, as moderate for SBP with renal events and for DBP with

cardiovascular events, and as low for DBP with all-cause mortality (Supplemental Table 24).

DISCUSSION

This dose-response meta-analysis illustrated the associations of BP levels with risks of all-cause mortality, cardiovascular, and renal outcomes in participants with type 2 diabetes. A J-shaped association was observed for SBP with all-cause mortality. J-shaped associations were also observed for SBP with cardiovascular events and for DBP with all-cause mortality, with flattened risks at lower BP levels. After excluding studies with participants

TABLE 2 Subgroup Analysis of Systolic Blood Pressure Level (per 10-mm Hg Increase) and Risks of Cardiovascular Events

Subgroup	No. of Cohorts	RR (95% CI)	I ² , %	τ ²	P _{between} ^a
All studies	29	1.06 (1.04-1.08)	88.8	0.0010	
Mean (or median) baseline age, y					0.137
<60	12	1.08 (1.04-1.13)	92.6	0.0018	
≥60	17	1.05 (1.02-1.07)	75.1	0.0007	
Female proportion, %					0.745
<50	13	1.06 (1.03-1.09)	91.3	0.0007	
≥50	16	1.07 (1.03-1.09)	76.8	0.0015	
Geographic location					0.391
Asia	14	1.08 (1.04-1.11)	72.2	0.0010	
North America	8	1.06 (1.01-1.11)	59.0	0.0003	
Europe	4	1.03 (0.96-1.11)	88.9	0.0011	
Others	3	1.04 (0.95-1.14)	38.8	0.0005	
Study design					0.343
Prospective	19	1.07 (1.04-1.10)	42.4	0.0011	
Retrospective	10	1.05 (1.01-1.09)	95.4	0.0010	
No. of participants					0.456
<500	8	1.09 (1.03-1.16)	0.0	0.0010	
500-999	6	1.08 (1.01-1.16)	42.8	0.0016	
1,000-4,999	7	1.05 (1.00-1.12)	0.0	0.0000	
≥5,000	8	1.05 (1.01-1.09)	96.5	0.0010	
Follow-up duration, y					<0.0001 ^b
<5	11	1.03 (0.99-1.06)	84.4	0.0004	
5-9.9	8	1.05 (1.04-1.07)	28.8	0.0000	
≥10	10	1.10 (1.08-1.11)	0.0	<0.0001	
Risk of bias					0.262
Low	23	1.05 (1.03-1.07)	90.8	0.0008	
Moderate	4	1.08 (0.98-1.20)	43.5	0.0021	
High or critical	2	1.26 (0.24-6.57)	73.0	0.0249	

^aP values for heterogeneity were estimated between subgroups by means of meta-regression analyses. Because multiple testing was performed, the risk of inflated type I error should be acknowledged. We presented CIs without multiplicity adjustment and did not provide P values for the pooled RRs. ^bSignificant P value after Bonferroni correction. Because the hypothesis test for meta-regression was run for 60 times altogether, the threshold was 0.05/60 after Bonferroni correction.

having baseline cardiovascular diseases or cancer, a monotonic trend between SBP and cardiovascular events and a J-shaped curve with a flattened lower end between SBP and all-cause mortality were observed, suggesting potential reverse causation. Meanwhile, BP levels were linearly or monotonically associated with all renal outcomes in this meta-analysis.

COMPARISON WITH TRIALS AND PREVIOUS META-ANALYSES. Few meta-analyses have explored the shapes of the associations between BP and clinical outcomes. Our findings aligned with a previous meta-analysis conducted by Prospective Studies Collaboration and Asia Pacific Cohort Studies Collaboration, which revealed a linear relationship between SBP and cardiovascular mortality in participants with type 2 diabetes.³³ Findings were also consistent with the current BP target

of <130/80 mm Hg for participants with type 2 diabetes recommended by both the American Diabetes Association³ and the American Heart Association.⁴

Recent RCTs support intensive BP lowering for the prevention of cardiovascular diseases. In the ESPRIT (Effects of Intensive Systolic Blood Pressure Lowering Treatment in Reducing Risk of Vascular Events) trial, which included 38.7% participants with type 2 diabetes, intensive treatment targeting SBP <120 mm Hg reduced the risk of major vascular events by 12% and all-cause mortality by 21% compared with standard treatment targeting SBP <140 mm Hg, with no evidence of heterogeneity by diabetes status.³⁴ In the BPROAD (Blood Pressure Control Target in Diabetes) trial, which included only type 2 diabetes patients, intensive treatment targeting SBP <120 mm Hg reduced the risk of major cardiovascular events by 21% compared with standard treatment targeting SBP <140 mm Hg, although no benefit was seen for all-cause mortality.³⁵ These landmark RCTs further underscore the importance of aggressive BP management in patients with diabetes, a high-risk population who may derive particular benefits from intensive control.

There is an ongoing debate regarding the U- or J-shaped associations of clinical outcomes with BP levels at the lower end. In the present meta-analysis, there was no clear evidence of a U-shaped association. Results from the main analysis revealed a nonsignificant trend toward increased risks of all-cause mortality at an SBP level of <120 mm Hg. However, potential reverse causation may play a part. Lower BP levels at the beginning of the study could also result from other serious diseases, such as cardiovascular diseases including heart failure or cancer. These baseline diseases could have seriously affected outcomes while at the same time lowering BP levels.^{36,37} Results from the sensitivity analysis after excluding studies with participants who had baseline cardiovascular diseases or cancer supported the existence of potential reverse causation between lower BP levels and risks of both cardiovascular events and all-cause mortality. Furthermore, time-varying confounding could also complicate interpretation. Changes in BP over time may be affected by disease progression, medication changes, or incident conditions, each of which can also directly influence outcomes. Therefore, findings from the present meta-analysis that low BP was not associated with an elevated risk of cardiovascular events or all-cause mortality compared with higher BP targets, suggest that previously reported harms of low SBP are most likely due to reverse causation and unmeasured confounding.

Regarding renal outcomes, some of the associations could not be assessed, particularly for DBP levels, owing to a limited number of cohorts. We observed a linear association between SBP levels and renal events, and a positive monotonic association between SBP levels and eGFR decline. These observational findings contrast with results from RCTs, where intensive BP-lowering treatment (eg, SBP target <120 mm Hg) increased the risk of eGFR decline, highlighting the difference in observational associations between SBP and renal function vs treatment effects by intensive BP intervention.

CLINICAL INTERPRETATION. Our findings suggest that, in people with type 2 diabetes, BP was linearly or monotonically associated with most cardiovascular and renal outcomes, and low SBP was not associated with an elevated risk of all-cause mortality. It is essential to emphasize that these observational patterns, derived from aggregate study-level data, suffer from residual confounding, reverse causation, etc. Sensitivity analysis excluding studies with baseline cardiovascular diseases or cancer was limited by the reliance on the reporting of baseline comorbidity data in the included studies. In addition, results from this analysis should not be used for treatment recommendations. Although our analysis offers valuable insights into population-level risk associations, clinical decision making should continue to be guided primarily by evidence from RCTs, which account for individual patient characteristics and infer causality. RCTs remain the criterion standard for establishing efficacy and safety of BP-lowering strategies. Our findings should be interpreted as generating hypotheses for further investigation rather than as justifying changes in clinical practice.

HETEROGENEITY. The considerable between-study heterogeneity observed in our pairwise meta-analyses, while extensively addressed methodology, warrants further contextual interpretation. Several factors may contribute to the observed heterogeneity. First, the heterogeneity is primarily statistical rather than methodologic. Although the included studies were highly consistent in the PICOS (population, intervention, comparison, outcomes and study) framework, statistical heterogeneity may still arise from variations in the level and precision of estimated effect sizes. Second, the large number of studies included, particularly for primary outcomes, also contributed to the between-study heterogeneity. As noted by Rücker et al,³⁸ a high I^2 value can reflect even minor variations in effect estimates when many studies are included. More importantly, the direction

TABLE 3 Subgroup Analysis of Systolic Blood Pressure Level (per 10-mm Hg Increase) and Risks of Renal Events

Subgroup	No. of cohorts	RR (95% CI)	I^2 , %	τ^2	P_{between}^a
All studies	16	1.22 (1.11-1.34)	91.8	0.0206	
Mean (or median) baseline age, y					0.293
<60	11	1.26 (1.12-1.42)	78.6	0.0155	
≥60	5	1.14 (0.91-1.43)	94.0	0.0251	
Female proportion, %					0.950
<50	8	1.23 (1.03-1.45)	94.4	0.0295	
≥50	8	1.22 (1.05-1.42)	81.3	0.0176	
Geographic location					0.662
Asia	7	1.29 (1.05-1.59)	85.5	0.0373	
North America	5	1.19 (1.07-1.32)	73.0	0.0027	
Europe	2	1.10 (0.32-3.85)	88.9	0.0174	
Others	2	1.27 (0.06-28.12)	95.4	0.1136	
Study design					0.988
Prospective	9	1.22 (1.05-1.43)	92.1	0.0287	
Retrospective	7	1.22 (1.04-1.44)	88.0	0.0163	
No. of participants					0.237
<500	2	1.29 (0.34-4.90)	22.8	0.0090	
500-999	7	1.32 (1.05-1.66)	88.3	0.0513	
1,000-4,999	2	1.19 (1.01-1.39)	0.0	0.0000	
≥5,000	5	1.12 (0.98-1.27)	97.0	0.0061	
Follow-up duration, y					0.965
<5	5	1.20 (0.92-1.57)	93.6	0.0354	
5-9.9	6	1.25 (0.95-1.64)	83.8	0.0332	
≥10	5	1.23 (1.03-1.48)	80.3	0.0155	
Risk of bias					0.808
Low	12	1.20 (1.06-1.36)	93.3	0.0231	
Moderate	3	1.30 (0.80-2.11)	89.4	0.0325	
High or critical	1	1.23 (1.08-1.40)	NA	NA	

^a P values for heterogeneity between subgroups were estimated by means of meta-regression analyses. Because multiple testing was performed, the risk of inflated type I error should be acknowledged. We presented CIs without multiplicity adjustment and did not provide P values for the pooled RRs.

NA = not applicable.

TABLE 4 Meta-Regression of Study-Level Characteristics on Associations of Systolic Blood Pressure Levels With Cardiovascular and Renal Events

Meta-Regression Variable	No. of Cohorts	Estimate	Lower Bound	Upper Bound	P Value	Adjusted R^2
Cardiovascular events						
Mean (or median) age	29	-0.0041	-0.0077	-0.0005	0.027	35.02
Female proportion	29	0.0004	-0.0012	0.0020	0.604	0.00
Follow-up duration	29	0.0107	0.0087	0.0127	<0.0001 ^b	99.54
No. of participants	29	0.0000	0.0000	0.0000	0.431	0.00
Risk of bias ^a	29	0.0487	-0.0003	0.0976	0.051	12.20
Renal events						
Mean (or median) age	16	-0.0111	-0.0311	0.0090	0.254	11.53
Female proportion	16	-0.0001	-0.0108	0.0106	0.991	0.00
Follow-up duration	16	0.0017	-0.0314	0.0348	0.912	0.00
No. of participants	16	0.0000	0.0000	0.0000	0.597	0.00
Risk of bias ^a	16	0.0355	-0.1344	0.2054	0.659	0.00

^aUsed as an ordinal variable with 3 levels: low, moderate, and high or critical. ^bSignificant P value after Bonferroni correction. Because the hypothesis test for meta-regression was run for 60 times altogether, the threshold was 0.05/60 after Bonferroni correction.

of associations was consistent across most studies. For example, among 26 studies examining SBP and cardiovascular events, 22 reported positive associations (15 significant). This consistency supports the potential appropriateness of data pooling despite quantitative heterogeneity. Meanwhile, the significant association between follow-up duration and SBP-related cardiovascular risk, accounting for 99.54% of heterogeneity, also reflects statistical heterogeneity arising from different effect sizes across studies. This finding aligns with our interpretation that the observed heterogeneity was most likely due to the quantitative differences in risk estimates while maintaining consistent and positive associations across all studies, supporting the validity of pooled estimates despite substantial statistical variation.

We contend that in the context of a large number of studies and a consistent direction of effects, the use of random-effects models and the Hartung-Knapp-Sidik-Jonkman method provides a conservative and appropriate approach to summarizing the evidence. The resulting estimates should be interpreted as describing a distribution of possible effects rather than a single uniform effect. While a high heterogeneity highlights the need for cautious interpretation, it does not invalidate the findings; rather, it underscores the complexity of BP-related risk in diabetes and the need for individualized clinical decision making supported by future trial evidence.

STUDY LIMITATIONS. There are several limitations to this study. First, as an aggregate-data meta-analysis rather than an individual-participant-data meta-analysis, the ability to adjust for confounding factors and conduct subgroup analysis was constrained. Second, despite the inclusion of studies under a consistent PICOS framework and a generally consistent direction of estimated effect sizes across cohorts, the large number of studies may have contributed to considerable heterogeneities, with significant variations observed across multiple outcomes. In addition, there might be regional differences in BP management or risk that cannot be fully addressed in the present analysis and further contributed to the heightened heterogeneities. Third, the reliance on single baseline BP measurements in most included studies represents an important limitation, because it fails to account for BP variabilities over time and may introduce measurement error or time-varying confounding. Fourth, the variability in composite endpoint definitions across studies also introduces clinical and statistical heterogeneity and increases uncertainty in the

pooled estimates. Although additional analyses on individual outcomes showed consistent effects, the inherent variability in endpoint composition remains a methodologic constraint.

CONCLUSIONS

This dose-response meta-analysis of 5.87 million participants with type 2 diabetes from 89 cohorts demonstrates the associations of BP levels and clinical outcomes. Findings suggest that in type 2 diabetes, BP was linearly or monotonically associated with most cardiovascular and renal outcomes. There was no clear evidence of a U-shaped association with all-cause mortality, especially after accounting for competing risks and potential reverse causation, suggesting that previously reported harms of low SBP in this population may be attributable to confounding. Future studies using large individual participant data are warranted, with considerations of sufficient lag time before onset of events and control for the confounding from reverse causation, time-varying variables, heterogeneities, etc to fully elucidate the associations between BP levels and clinical outcomes, particularly at very low BP levels.

DATA AVAILABILITY The data sets and codes used in this study are available from the corresponding authors on reasonable request.

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APPENDIX For supplemental material, please see the online version of this paper.